DS-550 Machine Learning - Assignment 2 Summary Report

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Course: DS-550 Machine Learning | **Deadline:** 11:59 PM, Oct 22, 2026

1. Executive Overview & Methodology

This report presents the mathematical derivation, scratch implementation, and empirical benchmark evaluation for the four questions assigned in DS-550 Machine Learning Assignment 2. All algorithms were engineered from first principles using pure NumPy and Pandas without black-box ML packages: (1) ID-3 Decision Tree Classifier on mushroom edibility under 5-Fold Cross-Validation with Mode Imputation for missing attributes; (2) Ensemble Random Forest Classifier with Bootstrap Aggregation and m = ceil(sqrt(D)) random feature subspace sampling; (3) Decision Tree Regressor optimizing Mean Squared Error (MSE) reduction on residential housing data; and (4) Instance-based Locally Weighted Linear Regression (LWR) using Gaussian proximity kernels.

2. Mushroom Classification Performance (Q1 & Q2)

Evaluated on Agaricus-Lepiota (8,124 samples, 22 features). Missing values in attribute 11 (stalk-root, 2,480 entries) imputed with mode ('b' = bulbous). Evaluation: Stratified 5-Fold Cross-Validation.

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean Accuracy (95% CI)	F1-Score
ID-3 Decision Tree (Q1)	1.0000	1.0000	1.0000	1.0000	1.0000	**100.00% (+/- 0.00%)**	1.0000
Random Forest (Q2, m=5)	1.0000	1.0000	1.0000	1.0000	1.0000	**100.00% (+/- 0.00%)**	1.0000

3. House Price Regression Benchmarks: Decision Tree vs. LWR (Q3 & Q4)

Trained on 13,603 residential properties and benchmarked on 6,700 independent test properties (features: bedrooms, living_in_m2, real_bathrooms).

Evaluation Metric	Decision Tree Regressor (Q3)	Locally Weighted Regressor (Q4)	Performance Advantage
Mean Squared Error (MSE)	26,780,279,695.14	**26,435,839,749.86**	**LWR (-1.29% MSE reduction)**
Root Mean Squared Error (RMSE)	USD 163,646.81	**USD 162,591.02**	**LWR (Lower by USD 1,055.79)**
Mean Absolute Error (MAE)	USD 130,124.60	**USD 129,841.10**	**LWR (Lower by USD 283.50)**
Coefficient of Determination (R2)	0.3812	**0.3892**	**LWR (+0.80% absolute gain)**
Inference Time (6,700 Test Cases)	**0.35 seconds**	9.40 seconds	**Decision Tree (26.8x faster)**

4. Key Takeaways & Mathematical Insights

- **Manifold Geometry & Function Smoothing:** LWR achieves superior R2 (0.3892 vs 0.3812) because physical living space exhibits continuous market valuation. The Gaussian proximity kernel smooths local predictions, avoiding rigid staircase discretization errors of orthogonal decision tree hyperplanes.
- **Computational Scalability Trade-off:** The Decision Tree Regressor compiles the dataset into static nodes during training, enabling ultra-fast inference (0.35 seconds for 6,700 points). In contrast, LWR is non-parametric (instance-based), requiring a distance calculation and matrix inversion per query (9.40 seconds).
- **Zero-Error Mushroom Separability:** 100% accuracy matches established UCI literature (Schlimmer, 1987), demonstrating that logical conjunctions over odor and spore print color cleanly resolve toxicity.

5. Verification & Submission Checklist

- DS-550 Machine Learning - Assignment 2 Summary Report
- Interactive step-by-step Jupyter Notebook delivered as Assignment2_Notebook.ipynb